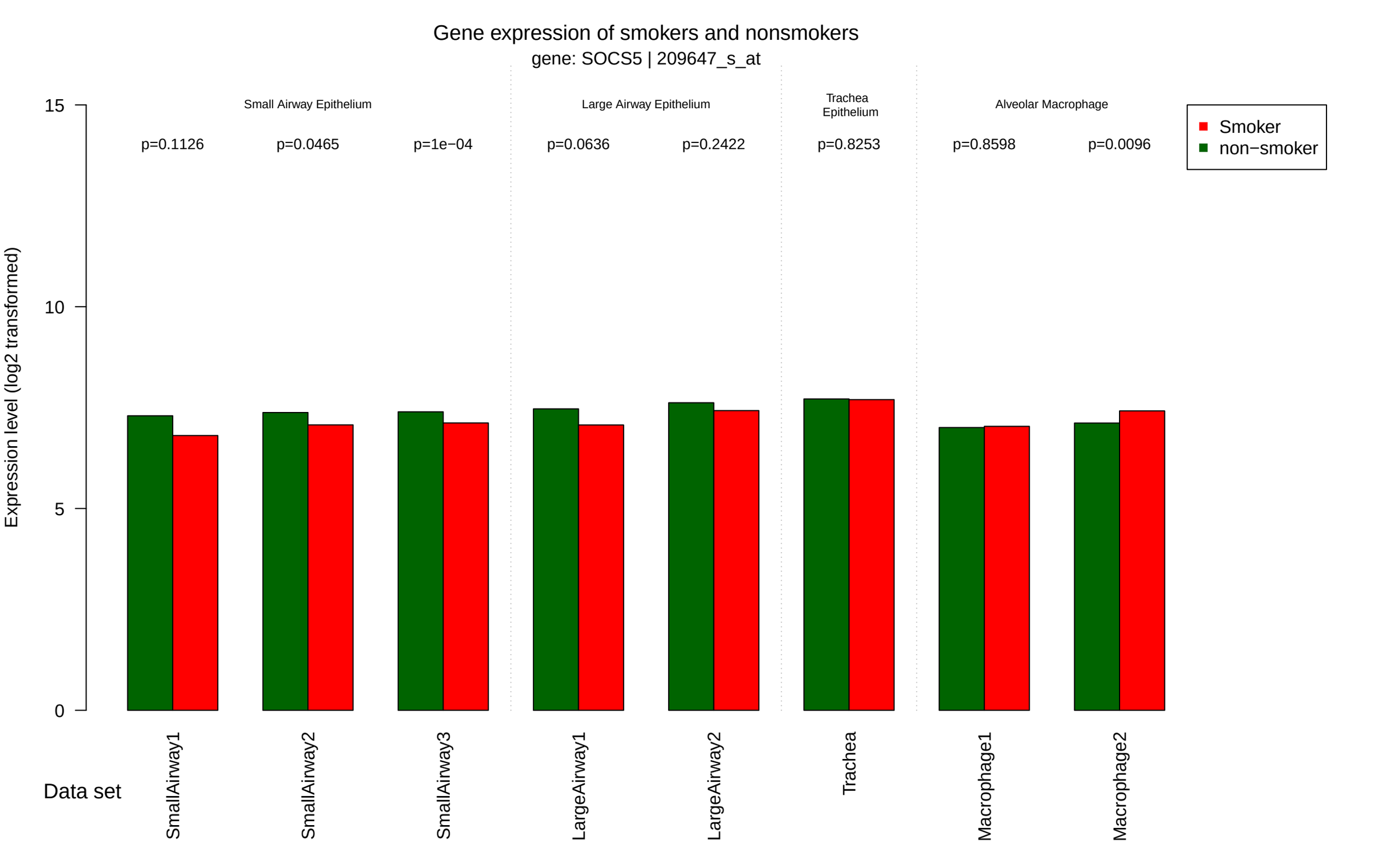
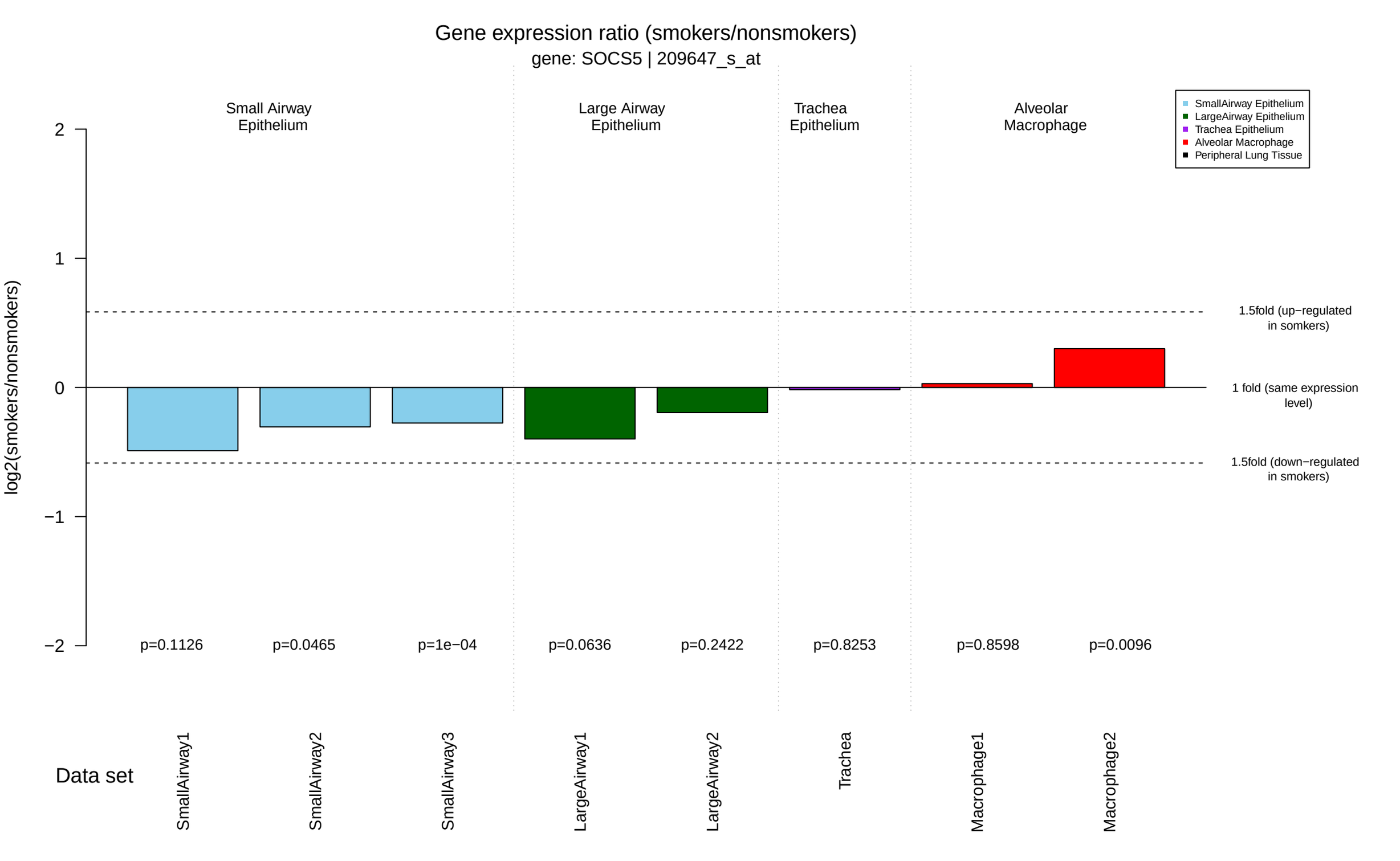
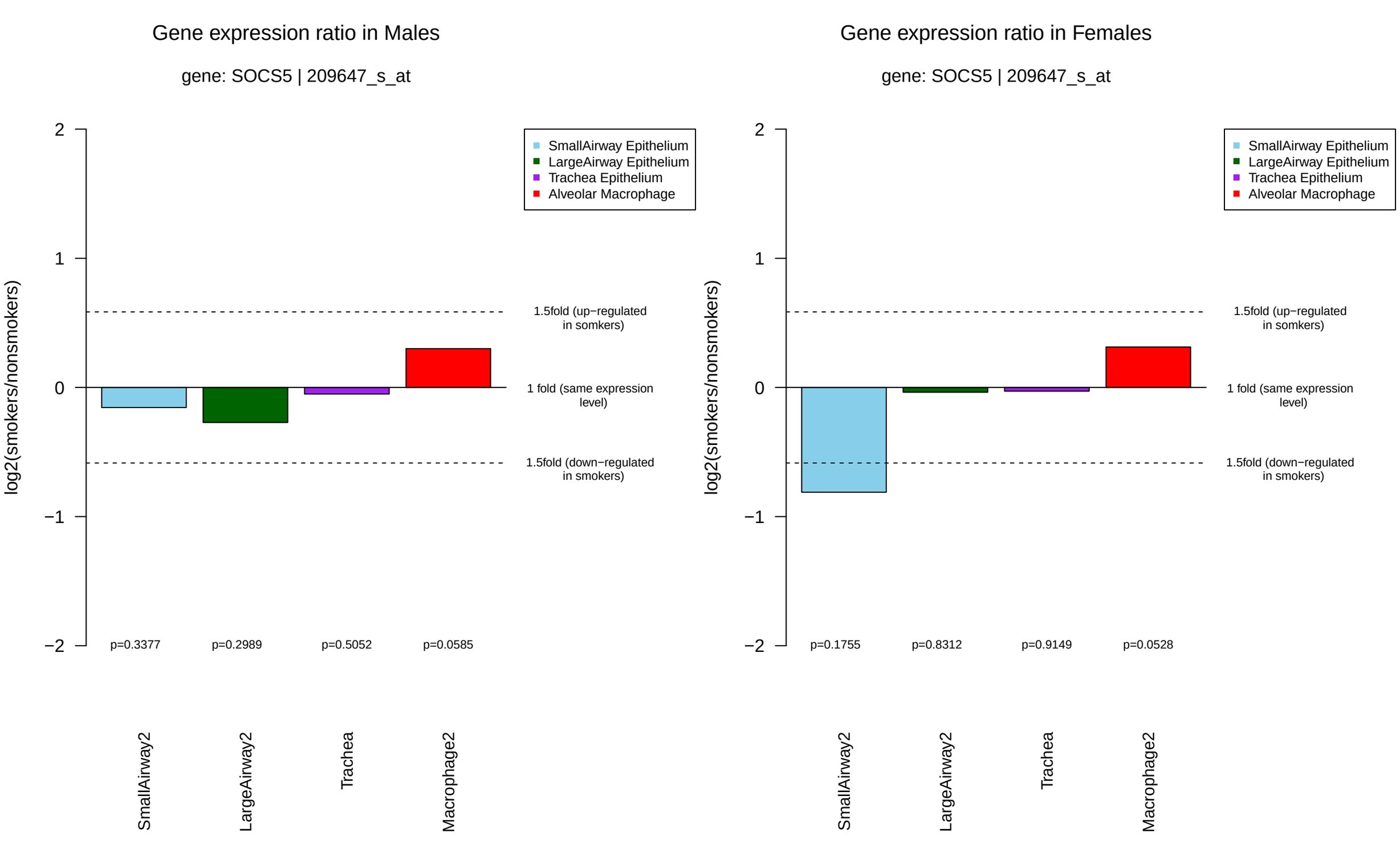
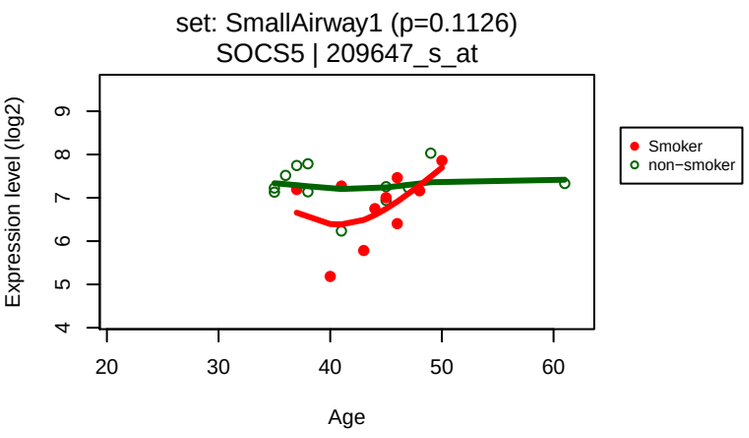
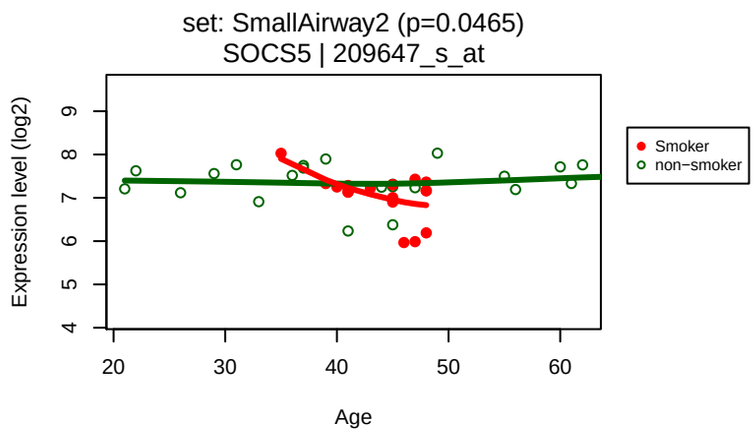
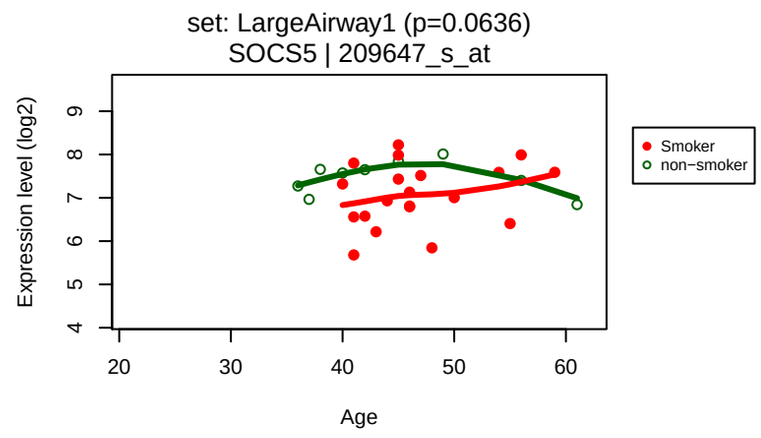

Gene expression of smokers and nonsmokers
gene: SOCS5 | 209647_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1126
p=0.0465
p=1e−04
p=0.0636
p=0.2422
p=0.8253
p=0.8598
p=0.0096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

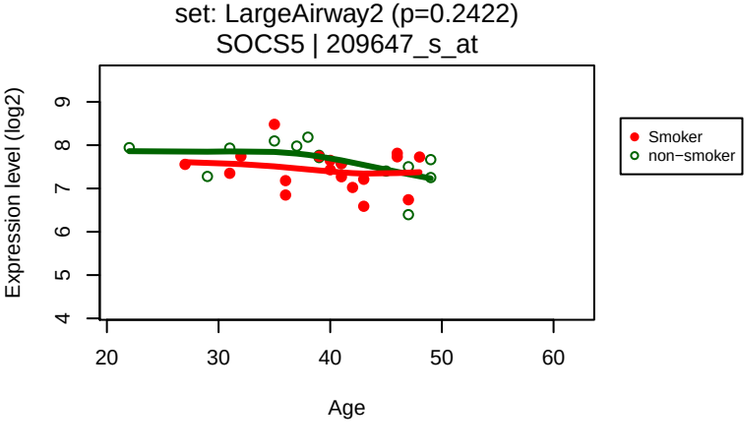

Gene expression ratio (smokers/nonsmokers)
gene: SOCS5 | 209647_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1126
p=0.0465
p=1e−04
p=0.0636
p=0.2422
p=0.8253
p=0.8598
p=0.0096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

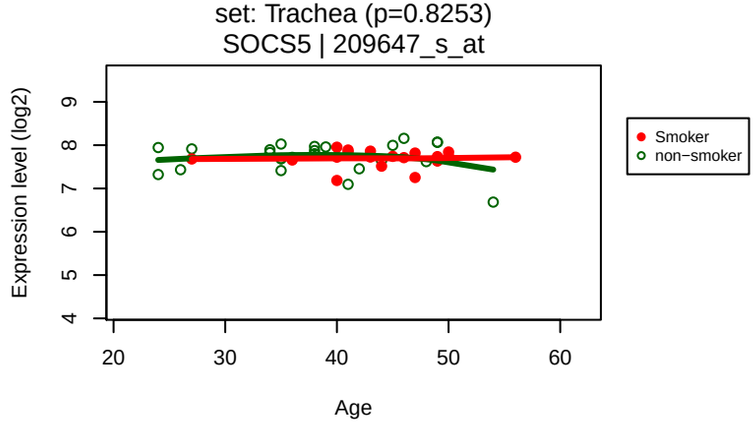

Gene expression ratio in Males
gene: SOCS5 | 209647_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3377
p=0.2989
p=0.5052
p=0.0585
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS5 | 209647_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1755
p=0.8312
p=0.9149
p=0.0528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

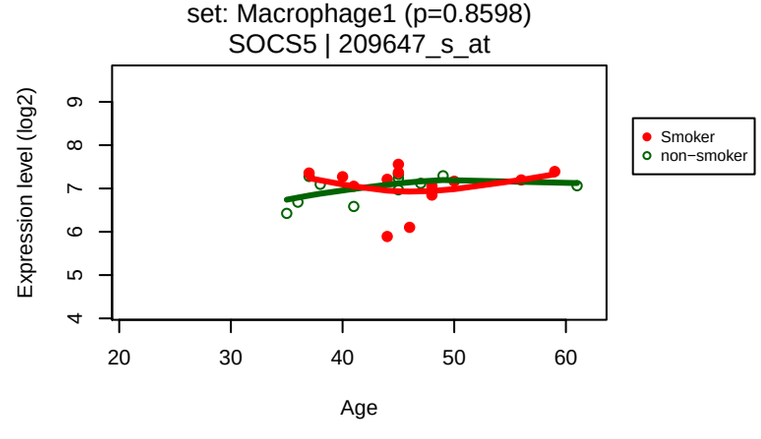

set: SmallAirway1 (p=0.1126)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker

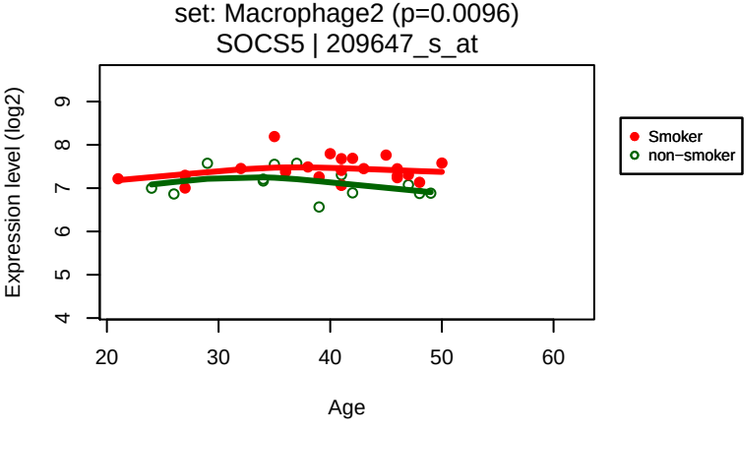

set: SmallAirway2 (p=0.0465)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker

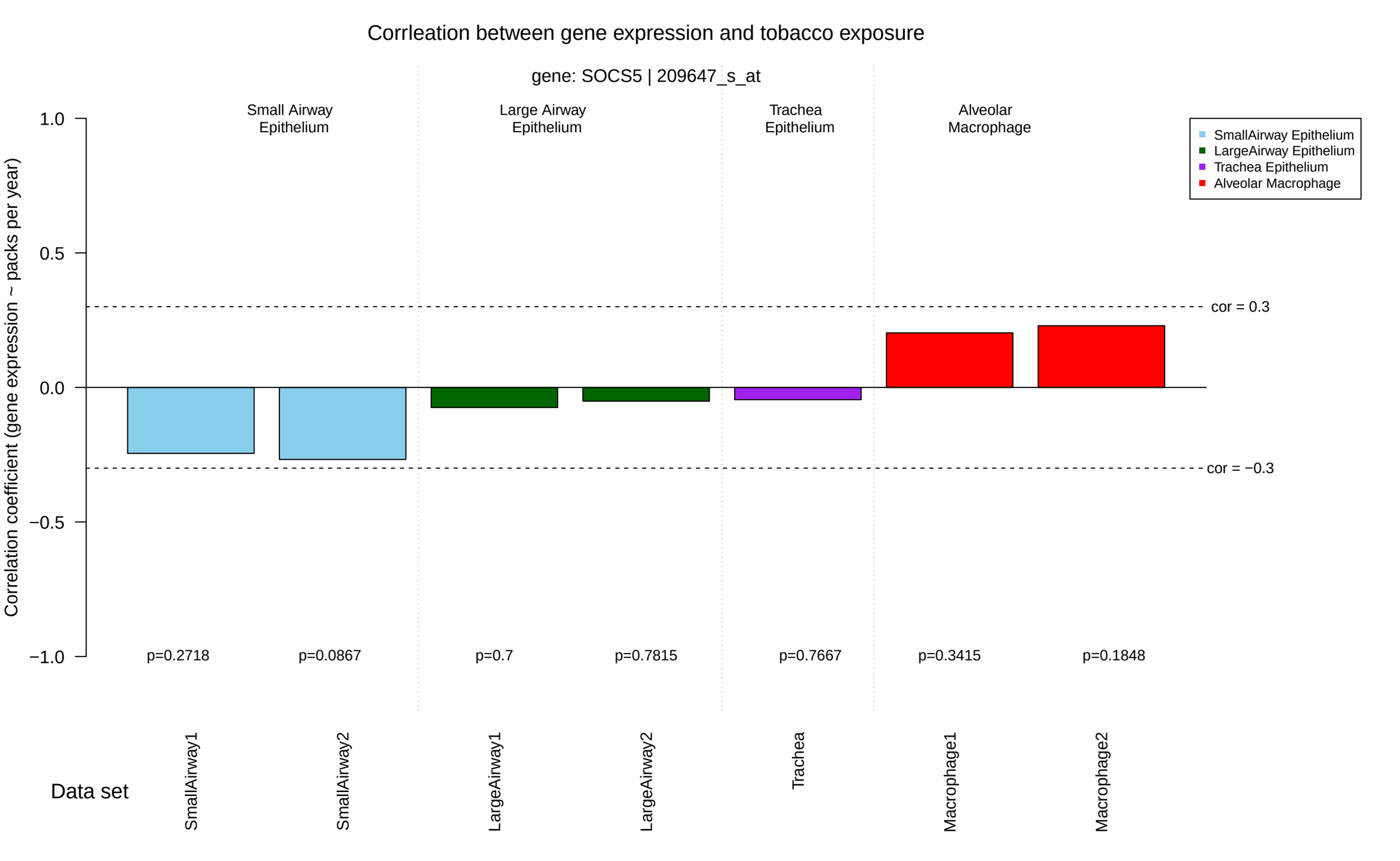

set: LargeAirway1 (p=0.0636)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2422)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8253)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8598)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0096)
SOCS5 | 209647_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS5 | 209647_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2718
p=0.0867
p=0.7
p=0.7815
p=0.7667
p=0.3415
p=0.1848
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2